

Tiny Bodies, Big Brains — Module 05: Action — Think About It

Part A: Family Discussion Questions

1. Why do babies crawl before they walk? (The brain builds movement skills from simple to complex).
2. What is the hardest movement you can do? (A cartwheel? Writing your name? Catching a ball?).
3. Why does practice make things easier? (The brain builds a faster "highway" for that movement).
4. What is "Proprioception"? (Your body's hidden sense of where it is in space).
5. Can you walk backward as easily as forward? Why not? (Less practice means less precise motor prediction).
6. Why do your hands shake when you try something new and tricky? (The brain is still learning the motor plan).
7. How did you learn to ride a bike, tie your shoes, or write your name? (Practice, mistakes, more practice!).
8. What movement can you do now that was impossible five years ago?
9. Why is it important to warm up before exercise? (Preparing the body systems for action).
10. How are your hands different from a dog's paws? (Thumbs! Fine motor skills let us write, draw, and build).

Part B: Hands-On Activities and Reflection

1. Try the "Balance Challenge": Stand on one foot with your eyes open. Time yourself. Now try with your eyes closed. Why is it harder? (Vision helps your balance system!).
2. Set up a simple obstacle course in your home (crawl under a table, jump over a pillow, spin three times). Time each family member. How does the body adapt to challenges?

3. Practice writing your name with your non-dominant hand. How does it feel? That is what it was like when you first learned to write — your brain had no motor plan for it yet!
4. Have a "Throwing Contest": Stand at a line and throw a soft ball at a target. Try ten times. Did you get better? Your motor prediction improved with each attempt.
5. Do the "Finger Dance": Have someone tap a rhythm on the table. Try to copy it. Start simple and add complexity. Your fingers are learning a new motor program!
6. Measure how far each family member can jump from a standing position. Write it down. Try again in a month. Did anyone improve? Growth upgrades the body system!
7. Try to balance a book on your head while walking across the room. What does your body have to do differently? (Adjust posture, slow down, engage core). That is motor adaptation!
8. Play "Freeze Dance": Dance when the music plays, freeze when it stops. Freezing requires inhibitory control — telling your body to STOP is a brain skill!
9. Draw the path your hand takes when you write the letter "S." Now watch a younger sibling or friend try the same letter. Their motor plan is still developing.
10. Before bed, stretch every part of your body from head to toe. Notice how each muscle feels. Thank your amazing body for all the movements it made today.